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MILITARY SPECIFICATION

GAGES, COMPRESSION AND COMPRESSION AND FIRING PRESSURE—DIESEL ENGINE CYLINDER

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 Scope. This specification covers high grade nonrecording instruments for gaging the compression and firing pressures within the cylinders of internal combustion diesel engines, and the special valve for attaching type I gages to the cylinder.

1.2 Classification. The gages shall be of the following types, as specified (see 6.1):

Type I —Maximum pressure, nonrecording, for compression and firing pressures.

Type II—Maximum compression pressure only; nonrecording.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiberboard.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-10377 — Box: Wood-Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case Waterproof.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 4910

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2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 — Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.).

3. REQUIREMENTS

3.1 **Preproduction model.** When specified (see 6.1) prior to production, a preproduction model shall be forwarded to a laboratory designated by the bureau or agency concerned for the tests specified in 4.2.

3.2 **Materials.** The materials shall be of the highest quality, noncorrosive when practicable, suitable for the purpose intended, and free of defects and imperfections that may affect the serviceability of the finished product. Transparent material, if used in the construction of the gage, shall remain clear and free from discoloration and shall be oil-resistant and shatterproof.

3.3 **Design.** The gage shall be so designed that no delicate adjustments are required, that it can be easily and quickly disassembled for servicing. The inertia and friction of parts actuated by the engine cylinder pressures shall be a minimum consistent with the principle of the design. If a Bourdon gage is used, the deflection of the Bourdon tube and pointer movement over the working range shall not exceed approximately ¾ of the movement required for maximum scale reading. If a helical spring is used, the deflection of the helical spring over the working range shall not cause a stress in excess of approximately ¾ the yield point of spring material. The dial diameter of Bourdon gages shall be not less than 2½ inches or more than 3½ inches.

3.4 **Resetting for successive use.** Means shall be provided for resetting the gage after each successive reading without disturbing the connection of the engine, and without use of tools for making adjustments or partial disassembly, so that successive readings or recordings may be taken quickly and accurately.

3.5 **Finish.** The finish of gage parts shall be in accordance with the best commercial practice. Metal parts not made of corrosion-resistant materials, shall be protected by a chemical or electro-chemical treatment to provide adequate protection against corrosion.

3.6 **Type I, maximum pressure, nonrecording, for compression and firing pressures.**

3.6.1 Gages shall be suitable for accurately determining peak compression pressures or firing pressures in engine cylinders equipped with gage valves or provided with gage valve connections. The gage shall have a working range of 0 to 2000 pounds per square inch (p.s.i.), the pressure indicating scale shall be marked in 20 p.s.i. increments and readable to 20 p.s.i. or less, with major graduations in 100 p.s.i. and numbered every 200 p.s.i. Gages shall be designed so that original calibration can be determined in the field without testing devices.

3.6.2 *Check valve.* The gas check valve, or pressure trap, shall be as light as practical, and of corrosion and heat resistant material. The check valve mechanism shall be readily accessible for cleaning or replacement of parts. The design and location of the valve shall not add appreciably to the clearance volume of the engine cylinder.

3.6.3 *Accessories and tools.* Each gage shall be furnished with the necessary tools for dis-

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assembly and cleaning of every part, one spare for each wearing part not requiring special skill for replacement, and one bottle of special lubricant and joint sealing compound if required.

3.6.4 *Connection.* The gage shall be fitted with a wing nut union connection for attachment to the engine gage valve. The wing nut thread shall be $1\frac{1}{16}$ inches truncated Whitworth, 10 threads per inch, and of class 2 fit in accordance with Handbook H28.

3.6.4.1 *Seal.* The dimensions of the cone seal between gage and valve shall be as follows:

Large diameter of cone — 0.786 to 0.788 inch.

Height of cone $^{31}\!/_{64} \pm \frac{1}{64}$ inches.

Included angle of cone — $7 \pm \frac{1}{2}$ degrees.

3.7 Type II, maximum compression pressure only; nonrecording.

3.7.1 Gages shall be suitable for accurately determining compression pressures of engines not provided with gage valves or gage valve connections. The gage dial shall have a working range of 50 to 1000 p.s.i., graduated 10 p.s.i. increments and readable to 10 p.s.i. or less. The dial or gage shall have major graduations at every 50 p.s.i. and shall be numbered at every 100 p.s.i.

3.7.2 *Connection to engine.* Each gage shall be furnished with adapters for attaching the gage to the engines, as specified (see 6.1). Adapters shall be the same general shape as the nozzle or injector being used in the engines, and be installed in the engines in the same manner.

3.7.3 *Check valve.* The check valve shall be as specified in 3.6.2.

3.7.4 *Accessories and tools.* Each gage shall be furnished with the necessary special tools for maintenance of the gage and for replacement of parts. One spare of each wearing part not requiring special skill for replacement and, if required, a container of special lubricant and joint sealing compound shall be furnished.

3.8 **Carrying case types I or II.** Each gage shall be furnished in a well made carrying case provided with a hinged lid, snap latches and carrying handle. The case shall be designed to hold the instrument, accessories, tools, repair parts and adapters provided, however, that if this would result in a case of unwieldy size, a separate case for any or all adapters can be supplied. The carrying case shall be marked in accordance with Standard MIL–STD–130.

3.9 Indicator valves for type I gages shall be as specified herein.

3.9.1 The valve body shall be forged of corrosion resisting steel having characteristics which will permit heat treating and hardening of the valve to the hardness indicated, and is resistant to corrosion and heat to which the valve is subjected.

(a) Valve body, 175 to 275 Brinell.

(b) $\frac{1}{2}$ inch pipe threads, 400 to 500 Brinell.

3.9.2 The valve spindle shall be of corrosion-resisting steel of heat and corrosion-resisting qualities. The spindle tip, or surface which seats in the valve body shall have a minimum hardness of 350 to 450 Brinell, to reduce wear, erosion, and leakage of the valve. The spindle shall be provided with a back-seat which will protect the thread from direct flow of hot gases when the valve is open. Spindle threads should be a national fine class 2 fit for proper alignment of the spindle tip in the body seat.

3.9.3 The valve design shall be such that the valve can be dismantled without removing the valve body from the engine. The cylinder end of the valve shall be provided with a $\frac{1}{2}$ inch national pipe thread. The gage side of the valve shall be provided with a thread and cone seat suitable for the gage connection specified in 3.6.4 and 3.6.4.1.

3.9.3.1 Flats shall be provided on the valve body for installing or removing the valve from the engine.

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3.9.3.2 A relief passage to the atmosphere shall be provided for venting accumulated gas in the valve or gage. The vent shall be automatically closed and opened as the valve is opened and closed.

3.9.3.3 A durable heat resistant, replaceable handle shall be provided for opening and closing the valve under operating conditions.

3.9.3.4 Thread protectors shall be provided for all exposed threads.

3.10 **Operating instructions.** The instruction booklet shall contain complete information necessary for operation and maintenance of the gage or gage valve with a list of replaceable parts. A copy of operating or maintenance instructions and parts list shall be included in each carrying case.

3.11 **Identification of product.** Gages shall be marked in accordance with Standard MIL–STD–130.

3.12 **Workmanship.** The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Preproduction.** Preproduction tests shall be conducted at a laboratory designated by the bureau or agency concerned. These tests shall consist of the tests specified in 4.2.1 and 4.2.2.

4.2.1 *Endurance.* The endurance test shall be made with the gage installed on one or more engines (operating at rated speed) for a period of 8 hours, during which period three consecutive pressure readings shall be taken every 30 minutes. At the conclusion of such tests, the gage shall not show evidence of heat damage or malfunctioning.

4.2.2 *Accuracy.* The gage shall be installed on various cylinders of a multi-cylinder engine or engines. Five consecutive pressure readings shall be taken at various loads and speeds. The pressures obtained with the gage shall be compared with the pressure indications of a gage of known accuracy. The variations in pressures with the comparative readings shall not exceed ± 2 percent.

4.3 **Inspection.**

4.3.1 *Sampling for acceptance inspection.*

4.3.1.1 *Inspection lot.* All gages of the same type and all indicator valves presented at one time shall be considered a lot for purposes of acceptance inspection.

4.3.1.2 *Sampling for visual and surface examination and tests.* A random sample of gages and gage valves shall be selected from each inspection lot by the inspector in accordance with table I and shall be subjected to the test specified in 4.3.2, and visual examination specified in 4.3.3. Acceptance or rejection of the lot shall be based on the sampling requirements of table I in accordance with Standard MIL–STD–105.

TABLE I. *Sampling for visual and surface examination and tests AQL (approx.) = 1.0 percent defective.*

Number of gages or gage valves in inspection lot	Number of gages or gage valves in sample	(Number of units nonconforming on visual examination or any test)	
		Acceptance number	Rejection number
15 and under	10	0	1
16 to 40	15	0	1
41 to 110	25	0	1
111 to 300	35	1	2
301 and over	50	1	2

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4.3.2 *Tests.* The gages or gage valves shall be surface tested to determine that they are of the same design and manufacture as those which have passed the preproduction testing.

4.3.3 *Visual and surface examination.* The inspector shall perform the following examination of sample gages selected in accordance with 4.3.1.2:

(a) Alignment and fit of parts.

(b) Finish.

(c) Interchangeability of parts.

(d) Hardness of valve body, and tightness of valve.

(e) Identification plate data.

(f) Accessories, tools, and repair parts.

(g) Adapters.

4.3.4 *Rejection.* Any gage or gage valve in the sample which contains one or more visual and surface defects or does not conform on any inspection test shall be rejected, and if the number of nonconforming gages or valves in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance inspection provided the contractor has removed all nonconforming material. The inspector shall again select and inspect samples from such resubmitted lots to verify compliance with this specification.

4.3.5 *Examination of preparation for delivery requirements.* Sample packages shall be selected and examined in accordance with the sampling provisions of Specification MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The carrying case with gage, accessories, tools and repair parts, secured and cushioned therein, shall be individually preserved and packaged in accordance with method 1A of Specification MIL-P-116. Contact preservative is not required.

5.1.2 *Level C.* The carrying case with gage, accessories, tools and repair parts, secured and cushioned therein, shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* The loaded carrying case, packaged as specified (see 6.1) shall be packed in overseas type wood cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood cleated paper overlaid, or wood cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636 (class 3), MIL-B-10377 or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with a minimum 1½ inch tape conforming to Specification PPP-T-76. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification..

5.2.2 *Level B.* The loaded carrying case, packaged as specified (see 6.1) shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood cleated plywood or wood cleated paper overlaid boxes or class 2 fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377, or PPP-B-636, respectively at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.3 *Level C.* The loaded carrying case, packaged as specified (see 6.1) shall be packed in containers of the type, size and kind commonly used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking required by the contract or order or herein, interior and exterior shipping container shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2).

(c) Whether a preproduction model is required (see 3.1).

(d) Adapters required for attaching the gage to the engine (see 3.7.2).

(e) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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